UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Unit 1 · Partial Differentiation · all sessions (Class 001 – Class 011) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)

S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

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A Formula sheet

2 Limit of a function of two variables

E2.1 · Limit of f(x, y) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

GIVES The value f approaches near (a, b).

APPLIES The same L along every path to (a, b).

NOTE For a polynomial (or any function continuous at the point) just substitute.

S001·9 · T001a 00:13:59

E2.2 · Two-path test (limit does not exist) ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x,y) \text{ does not exist}$$

GIVES A proof that the limit does not exist.

APPLIES Substitution gives $\frac{0}{0}$.

NOTE Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

3 Continuity of a function of two variables

E3.1 · Continuity at (a, b) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

GIVES The test for continuity.

APPLIES Three conditions: $f(a,b)$ defined, limit exists, both equal.

NOTE One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38

4 Partial derivatives: history, definition and notation

E4.1 · Partial derivatives from the definition ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h}$$

GIVES z_x (y held constant) and z_y (x held constant).

APPLIES When the limits exist.

NOTE Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

GIVES The equivalent symbols.

APPLIES $z = f(x, y)$.

NOTE The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

5 Second-order and mixed partial derivatives

E5.1 · Second-order partials ESA ★★

$$z_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}, \qquad z_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

GIVES Second derivatives in one variable.

APPLIES Differentiate z_x again in x (or z_y in y).

NOTE Same rules as ordinary second derivatives.

S001·16 · T001b 00:17:04

E5.2 · Mixed partials ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

GIVES The two crossed derivatives.

APPLIES Differentiate in one variable, then the other.

NOTE Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\tfrac{\partial^2 f}{\partial y\,\partial x} = \tfrac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES The order of differentiation does not matter.

APPLIES f_{xy} and f_{yx} continuous near the point.

NOTE Called “a very important theorem” in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

6 Higher-order partial derivatives

E6.1 · First-order partial derivatives

ESA ★★

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}$$

GIVES Rate of change of f in the x (or y) direction.

APPLIES Differentiate once; hold the other variable constant.

NOTE Two of them for a function of two variables.

S002·3 · T002 00:00:44

E6.2 · Second-order partial derivatives

ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES The four second-order partials of $f(x, y)$.

APPLIES Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.

NOTE Notation order varies between books. The two mixed partials are equal when continuous (E6.4).

ADDED FOR CLARITY S002·3 · T002 00:00:58

E6.3 · How many partial derivatives of order n?

ESA ★

2 variables: 2^n m variables: m^n

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n + 1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E6.4 · Equality of mixed partials (Clairaut)

ESA ★★

$$f_{xy} = f_{yx} \quad \Longleftrightarrow \quad \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES Order of differentiation does not matter.

APPLIES When f_{xy} and f_{yx} are continuous (true for every function in this unit's problems).

NOTE Verified in CW U1 P2 and P8.

TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:00:24 · T002 00:14:59 · CW U1 P2

7 The one-variable picture: derivative as slope

E7.1 · Derivative as slope (one variable) ESA ★

$$\left.\frac{dy}{dx}\right|_{x_0} = f'(x_0) = \text{slope of the tangent to } y = f(x) \text{ at } (x_0, f(x_0))$$

GIVES Slope of the tangent line.

APPLIES f differentiable at x_0.

S002·4 · T002 00:03:13

E7.2 · Tangent line to y = f(x) ESA ★

$$y - f(x_0) = f'(x_0)\,(x - x_0)$$

GIVES Equation of the tangent at x_0.

APPLIES f differentiable at x_0.

NOTE Point-slope form.

ADDED FOR CLARITY S002·4

9 First-order partials as slopes of tangents

E9.1 · ∂f/∂x as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x, y_0)\big)\right|_{x=x_0}$$

GIVES Slope of the tangent to the surface at $(x_0, y_0, f(x_0, y_0))$ that is parallel to the x-axis.

APPLIES Fix $y = y_0$ first; $f(x, y_0)$ depends only on x.

NOTE The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E9.2 · ∂f/∂y as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0, y)\big)\right|_{y=y_0}$$

GIVES Slope of the tangent to the surface that is parallel to the y-axis.

APPLIES Fix $x = x_0$ first; $f(x_0, y)$ "just depends on y".

NOTE The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E9.3 · The two tangent lines at P

ESA ★

$$x\text{-direction: } z - z_0 = f_x(x_0, y_0)\,(x - x_0),\;\; y = y_0$$
$$y\text{-direction: } z - z_0 = f_y(x_0, y_0)\,(y - y_0),\;\; x = x_0$$

GIVES Equations of the tangents whose slopes are f_x and f_y.

APPLIES $z_0 = f(x_0, y_0)$.

NOTE Point-slope form inside each plane section.

ADDED FOR CLARITY S002·7

10 Slope in any other direction: the gradient

E10.1 · Gradient

ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

GIVES Vector whose components are the slopes along the x- and y-axes.

APPLIES Taken at the point (x_0, y_0).

NOTE Covered fully in vector calculus.

S002·8 · T002 00:09:23

E10.2 · Slope along any direction (directional derivative)

ESA ★

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = a\,f_x + b\,f_y, \qquad \hat{u} = (a, b),\; a^2 + b^2 = 1$$

GIVES Slope of the tangent in the direction $\hat{u}$.

APPLIES $\hat{u}$ must be a unit vector.

NOTE $\hat{u} = \mathbf{i}$ gives f_x; $\hat{u} = \mathbf{j}$ gives f_y. The largest slope is $|\nabla f|$.

ADDED FOR CLARITY S002·8

11 Second-order partials: concavity and mixed partials

E11.1 · Concavity from second partials

ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

GIVES Shape of the surface along each axis direction.

APPLIES $f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.

NOTE Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19

E11.2 · Mixed partial as a change of slope ESA ★★

$$f_{xy} = \frac{\partial}{\partial y}\left(f_x\right)$$

GIVES How the x-slope f_x changes as you move in the y direction.

APPLIES Likewise $f_{yx} = \partial(f_y)/\partial x$: the change of the y-slope along x.

S002·9 · T002 00:11:45

E11.3 · Names of higher derivatives (motion) ESA ★

$$\frac{dy}{dt} = \text{velocity}, \quad \frac{d^2y}{dt^2} = \text{acceleration}, \quad \frac{d^3y}{dt^3} = \text{jerk}$$

GIVES Physical meaning when the independent variable is time.

APPLIES One-variable analogy only.

NOTE Names exist up to about the 6th derivative.

TRANSCRIPT ONLY T002 00:10:23 · T002 00:10:57

13 Composite functions and the one-variable chain rule

E13.1 · Chain rule, one variable ESA ★

$$\frac{d}{dx}\, f\big(g(h(x))\big) = f'\big(g(h(x))\big) \cdot g'\big(h(x)\big) \cdot h'(x)$$

GIVES Derivative of a composition: outer to inner, multiplied.

APPLIES Each function differentiable.

NOTE Example: $\dfrac{d}{dx} e^{\sin x^2} = e^{\sin x^2} \cos x^2 \cdot 2x$.

S003·4 · T003a 00:00:33

14 The total derivative rule

E14.1 · Total derivative rule ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

GIVES Rate of change of z with respect to the single variable t.

APPLIES $z = f(x, y)$, $x = x(t)$, $y = y(t)$.

NOTE Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E14.2 · Total derivative, three variables ESA ★★

$$\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}$$

GIVES One term per intermediate variable.

APPLIES $f = f(x, y, z)$, each of x, y, z a function of t.

NOTE Used in CW U1 P11 and in the triangle problem with x, y, θ.

ADDED FOR CLARITY S003·9 · CW U1 P11

15 The total differential: approximation and sensitivity

E15.1 · Total differential ESA ★★

$$dz = f_x(x, y)\, dx + f_y(x, y)\, dy$$

GIVES Approximate change in z for small changes dx, dy.

APPLIES $z = f(x, y)$ differentiable.

NOTE n variables: $dz = f_{x_1} dx_1 + f_{x_2} dx_2 + \cdots + f_{x_n} dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E15.2 · Approximating a nearby value ESA ★★

$$f(x_0 + dx,\, y_0 + dy) \approx f(x_0, y_0) + f_x(x_0, y_0)\, dx + f_y(x_0, y_0)\, dy$$

GIVES Estimated function value near a known point.

APPLIES dx, dy small.

NOTE Slide 003·8: $f(2.1, -3.03) \approx 6 + 1.3(0.1) + (-0.6)(-0.03) = 6.148$.

S003·8 · T003a 00:13:09

16 Rates of change problems

E16.1 · Constant quantity ESA ★★

$$A = \text{constant} \implies \frac{dA}{dt} = 0$$

GIVES The equation that fixes the unknown rate.

APPLIES "Remains constant" in the question.

S003·9 · T003a 00:16:32

17 The chain rule for composite functions

E17.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z = f(x, y)$, $x = g(u, v)$, $y = h(u, v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E17.2 · Chain rule, three intermediate variables ESA ★★★

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

GIVES One term per intermediate variable.

APPLIES $v = f(r, s, t)$, each of r, s, t a function of x, y, z.

NOTE A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E17.3 · Function of one intermediate variable ESA ★★

$$u = f(r),\ r = \sqrt{x^2 + y^2 + z^2} \implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x} = f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x, y, z only through r.

S004·12 · T004b 00:05:07

E17.4 · Polar coordinates: derivatives you reuse ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2 + y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r, \theta)$.

S004·13 · S004·15 · S004·17

E17.5 · Polar identity

ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u=u(x,y)$, $x=r\cos\theta$, $y=r\sin\theta$.

NOTE Proved in W17.16.

CW U1 P15

18 Implicit functions of two variables

E18.1 · Implicit differentiation, two variables

ESA ★★

$$f(x,y)=c \implies \frac{dy}{dx}=-\frac{f_x}{f_y}, \qquad f_y\neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y\neq 0$ at the point.

NOTE From $df=f_x\,dx+f_y\,dy=0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E18.2 · Tangent line at a point

ESA ★★

$$y-y_0=m\,(x-x_0), \qquad m=\left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

GIVES Equation of the tangent to the curve at (x_0,y_0).

APPLIES (x_0,y_0) on the curve.

NOTE Through the origin: $y=mx$.

S004·7 · T004a 00:14:25

E18.3 · Derivative of a constant power and a variable power

ESA ★★

$$\frac{d}{dx}a^x=a^x\ln a, \qquad \frac{d}{dx}x^n=n\,x^{n-1}$$

GIVES The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.

APPLIES In a partial derivative, the other variable is a constant.

NOTE $\ln$ needs a positive argument: $\cos x>0$, $\sin y>0$.

S004·8 · T004a 00:16:43

19 Implicit functions of three variables

E19.1 · Implicit differentiation, three variables

ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

20 Homogeneous functions

E20.1 · Homogeneous of degree n

ESA ★★

$$u=x^n g\!\left(\frac{y}{x}\right)=y^n h\!\left(\frac{x}{y}\right) \quad\Longleftrightarrow\quad u(tx,ty)=t^n\,u(x,y)$$

GIVES The definition and the equivalent scaling test.

APPLIES Two variables; with three, use $x^n g(y/x, z/x)$.

NOTE The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E20.2 · Degree of a quotient and of a product

ESA ★★

$$\deg\frac{P}{Q}=\deg P-\deg Q, \qquad \deg(PQ)=\deg P+\deg Q$$

GIVES Degree of combinations of homogeneous functions.

APPLIES P, Q homogeneous.

NOTE $\frac{\sqrt{x}+\sqrt{y}}{x+y}$: $\frac{1}{2}-1=-\frac{1}{2}$. A ratio such as $\sin^{-1}(y/x)$ has degree 0.

ADDED FOR CLARITY T005 00:07:50

21 Euler's theorem

E21.1 · Euler's theorem, two variables

ESA ★★★

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=n\,u$$

GIVES The scaled sum of first partials.

APPLIES u homogeneous of degree n.

NOTE Compact form: $xu_x+yu_y=nu$.

S005·6 · T005 00:08:57

E21.2 · Euler's theorem, three variables ESA ★★

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=n\,u$$

GIVES The same result with one term per variable.

APPLIES $u=f(x,y,z)$ homogeneous of degree n.

NOTE Extends to any number of variables.

S005·6 · T005 00:14:28

22 Second and higher order: the corollary

E22.1 · Corollary: second order ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=n(n-1)\,u$$

GIVES The scaled sum of second partials.

APPLIES u homogeneous of degree n with continuous second partials.

NOTE Coefficients $1,2,1$ as in $(x+y)^2$.

S005·6 · T005 00:12:21

E22.2 · Third order ESA ★

$$x^3u_{xxx}+3x^2y\,u_{xxy}+3xy^2u_{xyy}+y^3u_{yyy}=n(n-1)(n-2)\,u$$

GIVES The pattern continues with binomial coefficients.

APPLIES Third partials continuous.

NOTE The lecture states the right-hand side; the coefficients 1, 3, 3, 1 are added for clarity.

TRANSCRIPT ONLY ADDED FOR CLARITY T005 00:12:43 · T005 00:14:12

E22.3 · First plus second order ESA ★★★

$$x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}+x\,u_x+y\,u_y=n(n-1)u+nu=n^2u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u=u_1+u_2$ apply to each part.

NOTE Degrees 3 and -3 both give 9: so $9u$ for $x^3\sin^{-1}\frac{y}{x}+y^{-3}\tan^{-1}\frac{x}{y}$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

23 Extension: when f(u) is homogeneous

E23.1 · Extension, first order

ESA ★★★

$$f(u) \text{ homogeneous of degree } n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES Euler's combination for a function that is not itself homogeneous.

APPLIES Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).

NOTE $\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.

S005·7

E23.2 · Extension, second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order combination for the extension.

APPLIES $g(u) = n\,f(u)/f'(u)$ from E23.1.

NOTE Differentiate g with respect to u.

S005·7

25 Testing homogeneity: the scaling test

E25.1 · Scaling test

ESA ★★

$$u(\lambda x, \lambda y) = \lambda^n\,u(x,y) \implies u \text{ homogeneous of degree } n$$

GIVES Whether u is homogeneous, and its degree.

APPLIES Any function; with three variables scale z too.

NOTE If no power λ^n factors out, u is not homogeneous.

S006·5 · T006b 00:00:52

E25.2 · Degree of a quotient

ESA ★★

$$\deg\frac{P}{Q} = \deg P - \deg Q$$

GIVES The degree without writing out λ.

APPLIES P, Q homogeneous.

NOTE $\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $1-4=-3$.

ADDED FOR CLARITY S006·4 · T006a 00:06:10

26 Verifying Euler's theorem directly

E26.1 · Euler's theorem

ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

GIVES The first-order Euler sum.

APPLIES u homogeneous of degree n; add $z\,u_z$ for three variables.

NOTE To verify, compute the left side by differentiating and show it equals nu.

S006·3 · S006·10 · T006a 00:00:00

27 When u itself is homogeneous: first and second order

E27.1 · Second order

ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$$

GIVES The second-order Euler sum.

APPLIES u homogeneous of degree n.

NOTE $n = -1$ gives $2u$; $n = -\frac{2}{3}$ gives $\frac{10}{9}u$.

S006·11 · T006b 00:07:23

E27.2 · First plus second order

ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n^2u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u_1 + u_2$ use $n_1^2u_1 + n_2^2u_2$.

NOTE $n(n-1) + n = n^2$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

28 When f(u) is homogeneous: the extension

E28.1 · Extension, first order

ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES The Euler sum when u is not homogeneous but $f(u)$ is.

APPLIES $f(u)$ homogeneous of degree n; add $z\,u_z$ for three variables.

NOTE $\sin \to n\tan u$, $\tan \to \frac{n}{2}\sin 2u$, $e^u \to n$, $\log \to nu\log u$.

S006·13 · S006·15 · S006·17 · T006b 00:09:08

E28.2 · Extension, second order

ESA ★★★

$$x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order sum for the extension.

APPLIES g from E28.1; differentiate g with respect to u.

NOTE If g is a constant (the $\log$ case), the answer is $-g$.

S006·15 · S006·17 · T006b 00:10:59

E28.3 · Deriving the extension by the chain rule

ESA ★★

$$F = f(u):\ xF_x + yF_y = nF, \quad F_x = f'(u)\,u_x \ \Rightarrow\ f'(u)\,(xu_x + yu_y) = nf(u)$$

GIVES Where E28.1 comes from; T006a works slide 006·4 this way.

APPLIES When you prefer not to quote E28.1.

NOTE On slide 006·4: $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$; divide by $\cos u$.

S006·4 · T006a 00:07:33 · T006a 00:08:22

30 Taylor's and Maclaurin's series in one variable

E30.1 · Taylor's series, one variable

ESA ★

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \ldots$$

GIVES A polynomial form of f near $x = a$.

APPLIES f has derivatives of every order at a and the series converges to f.

NOTE Slide 007·4 writes $f^n(a)$ for the n-th derivative $f^{(n)}(a)$.

S007·4 · T007 00:04:56

E30.2 · Maclaurin's series, one variable

ESA ★

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots$$

GIVES Taylor's series about the origin.

APPLIES f and all its derivatives finite at 0 (not, e.g., $1/x$ or $\log x$).

NOTE "About a point $x = a$: Taylor; about the origin: Maclaurin."

S007·4 · T007 00:11:17

E30.3 · Exponential and logarithm

ESA ★★

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (x \in \mathbb{R})$$
$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (|x| < 1)$$

GIVES Two standard series.

APPLIES As stated.

NOTE e^{-x}: replace x by $-x$.

S007·5 · T007 00:07:19 · T007 00:08:39

E30.4 · Binomial and geometric series

ESA ★

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$
$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \qquad (|x| < 1)$$

GIVES Series for powers.

APPLIES $|x| < 1$ (any n; for a positive integer n the series stops).

NOTE The geometric series is said in the lecture, not on the slide.

S007·5 · T007 00:07:37

E30.5 · Trigonometric and hyperbolic

ESA ★★

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$
$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

GIVES Odd and even standard series.

APPLIES All real x.

NOTE Slide 007·5 prints $\cos x = 1 - \frac{x^2}{2} + \dots$ ($2! = 2$).

S007·5

E30.6 · Inverse trigonometric

ESA ★

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \dots, \qquad \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

GIVES Series for the inverse functions.

APPLIES $-1 \le x \le 1$.

NOTE Slide 007·5 prints $\tan^{-1} x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$; the denominators are 3 and 5, not $3!$ and $5!$.

S007·5

31 Taylor's expansion of a function of two variables

E31.1 · Taylor's series, two variables (to second order)

ESA ★★★

$$f(x,y) = f + \frac{1}{1!}\{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots, \quad h = x-a,\ k = y-b$$

GIVES A polynomial form of f near (a,b).

APPLIES All derivatives evaluated at (a,b), all finite there.

NOTE h, k are shorthand added for clarity; the slide writes $(x-a)$, $(y-b)$.

S007·9 · T007 00:18:07 · T007 00:18:45

E31.2 · Third-order term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms needed for "up to third degree".

APPLIES Add to E31.1 (or E32.1 with $h = x$, $k = y$).

NOTE Not on the slides; asked in four of six ESA Taylor questions.

ADDED FOR CLARITY ESA Dec-23 1b · ESA Jan-24 s1 1b · ESA Jan-24 s2 1b · ESA Jun-24 1c

E31.3 · Operator form

ESA ★★

$$f(a+h, b+k) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^n f(a,b)$$

GIVES Why the coefficients are binomial: 1, 2, 1 and 1, 3, 3, 1.

APPLIES Mixed partials equal ($f_{xy} = f_{yx}$).

NOTE Expand $(h\partial_x + k\partial_y)^n$ like $(h+k)^n$.

ADDED FOR CLARITY T007 00:18:26

32 Maclaurin's expansion of a function of two variables

E32.1 · Maclaurin's series, two variables

ESA ★★★

$$f(x,y) = f(0,0) + \frac{1}{1!}\{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$$

GIVES A polynomial form of f near the origin.

APPLIES Derivatives at $(0,0)$, all finite there.

NOTE Same as E31.1 with $a = b = 0$; add E31.2 with $h = x$, $k = y$ for third degree.

S007·10 · T007 00:19:38

33 Taylor expansion about a point (a, b)

E33.1 · Taylor's series about (a, b), to second order

ESA ★★★

$$f(x,y) = f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots, \quad h = x - a,\ k = y - b$$

GIVES The polynomial to substitute into.

APPLIES All derivatives evaluated at (a,b).

NOTE h, k are shorthand added for clarity.

S008·3 · S008·14 · T008 00:00:37 · T008 00:01:25

E33.2 · Third-degree term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms for “up to third degree”.

APPLIES Add to E33.1 or E34.1.

NOTE Coefficients 1, 3, 3, 1; asked in four of six ESA questions.

S008·10 · ESA Dec-23 1b · ESA Jun-24 1c

34 Maclaurin expansion about the origin

E34.1 · Maclaurin's series (about the origin)

ESA ★★★

$$f(x,y) = f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$$

GIVES The polynomial about $(0,0)$.

APPLIES Derivatives at $(0,0)$.

NOTE Slide 008·10 calls this “Taylor's series about (0,0)”; it is the same thing.

S008·10 · S008·12 · T008 00:06:58

E34.2 · One-combination check ESA ★★

$$f = g(t),\ t = px + qy \implies f(x,y) = g(0) + g'(0)\,t + \frac{g''(0)}{2!}t^2 + \frac{g'''(0)}{3!}t^3 + \dots$$

GIVES The same expansion in one line.

APPLIES f depends on x and y only through $px + qy$.

NOTE $\log(2x + y + 1) = t - \frac{t^2}{2} + \frac{t^3}{3}$ with $t = 2x + y$; $e^{x-y} = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}$ with $t = x - y$.

ADDED FOR CLARITY S008·6 · S008·11

35 Approximate values from a Taylor polynomial

E35.1 · Approximation by a Taylor polynomial ESA ★★

$$f(a+h, b+k) \approx f(a,b) + hf_x + kf_y + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$$

GIVES A numerical estimate.

APPLIES h, k small; (a,b) a point where f and its derivatives are easy.

NOTE Choose (a,b) with "nice" values: $(1,-1)$ for $(0.9,-1.2)$.

S008·7 · S008·9 · T008 00:12:10

37 Definitions: relative maximum, minimum and extreme value

E37.1 · Sign of the increment ESA ★

$$\Delta = f(a+h, b+k) - f(a,b) \quad \begin{cases} < 0 \text{ for all small } h,k: & \text{maximum} \\ > 0 \text{ for all small } h,k: & \text{minimum} \end{cases}$$

GIVES The definition of an extreme value.

APPLIES Used to settle the doubtful case $rt - s^2 = 0$.

NOTE Same sign in every direction.

S009·6 · T009 00:05:23

38 Conditions and the working rule

E38.1 · Stationary points (necessary condition) ESA ★★★

$$f_x = \frac{\partial f}{\partial x} = 0, \qquad f_y = \frac{\partial f}{\partial y} = 0$$

GIVES The candidates $(a, b), (c, d), \ldots$ for extreme values.

APPLIES f differentiable; solve the two equations simultaneously.

NOTE Two-variable version of $\frac{dy}{dx} = 0$.

S009·7 · S009·8 · T009 00:06:19

E38.2 · Second derivatives used in the test ESA ★★★

$$r = \frac{\partial^2 f}{\partial x^2}, \qquad s = \frac{\partial^2 f}{\partial x\, \partial y}, \qquad t = \frac{\partial^2 f}{\partial y^2}$$

GIVES The three numbers at each stationary point.

APPLIES Evaluate at each stationary point separately.

NOTE $s = f_{xy} = f_{yx}$ (Clairaut).

S009·7 · S009·8 · T009 00:07:04

E38.3 · Hessian determinant ESA ★★★

$$rt - s^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

GIVES Whether the stationary point is an extremum.

APPLIES At a stationary point.

NOTE Positive: extremum (decide by r). Negative: saddle. Zero: doubtful.

S009·7 · T009 00:07:37

E38.4 · The test (working rule, steps 3–6) ESA ★★★

$$rt - s^2 > 0,\ r < 0 \ \Rightarrow\ \text{maximum}; \qquad rt - s^2 > 0,\ r > 0 \ \Rightarrow\ \text{minimum}$$
$$rt - s^2 < 0 \ \Rightarrow\ \text{saddle point}; \qquad rt - s^2 = 0 \ \Rightarrow\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES After E38.1–E38.3.

NOTE $r < 0$ plays the role of $\frac{d^2y}{dx^2} < 0$ in one variable. When $rt - s^2 > 0$, r and t have the same sign, so t can be used instead.

S009·8 · S009·9 · T009 00:11:32

39 The method, and the ESA cubic functions

E39.1 · Stationary points

ESA ★★★

$$f_x = 0, \qquad f_y = 0 \quad \text{(solve simultaneously)}$$

GIVES All candidate points.

APPLIES Every problem, first step.

NOTE Factorise: $6xy - 30y = 6y(x-5) = 0$ gives the cases $y = 0$ or $x = 5$.

S010·7 · T010b 00:01:23 · T010b 00:01:40

E39.2 · The test

ESA ★★★

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}$$

$$rt - s^2 > 0:\ r < 0 \text{ max},\ r > 0 \text{ min}; \qquad rt - s^2 < 0:\ \text{saddle}; \qquad rt - s^2 = 0:\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES At each point separately.

NOTE Restated from class 009.

S010·8 · T010b 00:03:12 · T010b 00:04:02

E39.3 · Hessian of the ESA cubic

ESA ★★★

$$f = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x: \qquad rt - s^2 = 36\left[(x-5)^2 - y^2\right]$$

GIVES $rt - s^2$ at every point at once.

APPLIES Dec-23 1c, Jan-24 s1 1a.

NOTE $r = t = 6x - 30$, $s = 6y$. Slide 010·8 drops the 36 on y^2.

S010·8 · T010b 00:03:30

41 Applied problems: eliminate one variable first

E41.1 · Eliminating with a sphere

ESA ★★

$$x^2 + y^2 + z^2 = a^2 \ \Rightarrow\ z^2 = a^2 - x^2 - y^2, \qquad T = kxyz^2 = kxy(a^2 - x^2 - y^2)$$

GIVES A function of x, y only.

APPLIES Quantity contains z^2 and the point lies on a sphere.

NOTE Unit sphere: $a = 1$.

S010·3 · S010·9 · T010a 00:01:00

E41.2 · Open box of volume V

ESA ★★

$$A = xy + 2yz + 2zx, \quad xyz = V \;\Rightarrow\; A = xy + \frac{2V}{x} + \frac{2V}{y}$$

GIVES Surface area (metal used) of a box with no top, base $x \times y$.

APPLIES Least material for a given volume.

NOTE Minimum at $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$ (CW P34). The slide's $A = 2xy + 2yz + zx$ is not the area of an open box.

ADDED FOR CLARITY S010·5 · CW U1 P34

42 Lagrange's method: the auxiliary function and the procedure

E42.1 · Auxiliary (Lagrange) function

ESA ★★★

$$F(x, y, z) = f(x, y, z) + \lambda\, \phi(x, y, z)$$

GIVES One function whose stationary points are the constrained stationary points.

APPLIES Optimise f subject to $\phi(x, y, z) = 0$.

NOTE Write the constraint with everything on one side: $x^2 + y^2 + z^2 - r^2$, $xy + 2yz + 2zx - 432$.

S011·3 · T011a 00:01:08

E42.2 · The four equations

ESA ★★★

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi = 0$$

GIVES x, y, z and λ (four unknowns, four equations).

APPLIES Always; with two variables drop F_z (three equations).

NOTE Equivalently $\nabla f = -\lambda \nabla \phi$: the gradients are parallel (added for clarity).

S011·3 · T011a 00:02:17 · T011a 00:02:30

E42.3 · Solving trick: make λ the subject

ESA ★★

$$\lambda = -\frac{f_x}{\phi_x} = -\frac{f_y}{\phi_y} = -\frac{f_z}{\phi_z}$$

GIVES Relations between x, y, z without λ.

APPLIES When $\phi_x, \phi_y, \phi_z \neq 0$.

NOTE Or multiply the equations by x, y, z and subtract (slides 011·6, 011·11).

S011·13 · S011·14 · T011b 00:07:24

43 Geometric problems: inscribed solids, nearest and farthest points

E43.1 · Largest box in a sphere and in an ellipsoid ESA ★★★

$$x^2+y^2+z^2=r^2:\ x=y=z=\tfrac{r}{\sqrt{3}},\ V_{\max}=\tfrac{8r^3}{3\sqrt{3}}$$
$$\tfrac{x^2}{a^2}+\tfrac{y^2}{b^2}+\tfrac{z^2}{c^2}=1:\ x=\tfrac{a}{\sqrt{3}},\ y=\tfrac{b}{\sqrt{3}},\ z=\tfrac{c}{\sqrt{3}},\ V_{\max}=\tfrac{8abc}{3\sqrt{3}}$$

GIVES Half-edges and the largest volume $V=8xyz$.

APPLIES Box centred at the origin, edges $2x, 2y, 2z$ parallel to the axes.

NOTE The sphere is the ellipsoid with $a=b=c=r$: the box is a cube.

S011·5 · ESA May-24 1c · ESA Jan-24 s3 1a · ESA Jan-24 s4 1a

E43.2 · Distance problems: use the square ESA ★★★

$$f=x^2+y^2+z^2\ \ (\text{origin}),\qquad f=(x-x_0)^2+(y-y_0)^2+(z-z_0)^2$$

GIVES The squared distance, which has its extrema at the same points as the distance.

APPLIES Nearest or farthest point on a curve or surface.

NOTE Take the square root at the end.

ADDED FOR CLARITY S011·8 · ESA Jul-23 1c · CW U1 P36

B Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	x, y independent, z dependent	Class 001: T1
D	Domain	Set of ordered pairs where f is defined	Class 001: T1
L	Limit	Common value along every path	Class 001: E2.1
(a, b)	Point of approach	Where the limit or continuity is tested	Class 001: E2.1, E3.1
m	Slope of a path	Path $y = mx$ in the two-path test	Class 001: E2.2, W2.2
∂	Partial ("dou")	Partial derivative sign; not δ, not d	Class 001: E4.1–E5.3
h	Increment	Small change in one variable in the definition	Class 001: E4.1
$z_x,\ f_x$	First partial in x	$\frac{\partial z}{\partial x}$ with y constant	Class 001: E4.2
$z_{xx},\ z_{yy}$	Second partials	$\frac{\partial^2 z}{\partial x^2},\ \frac{\partial^2 z}{\partial y^2}$	Class 001: E5.1
z_{xy}	Mixed partial	x first, then y; $= \frac{\partial^2 z}{\partial y \partial x}$	Class 001: E5.2, E5.3
$f,\ z$	Function / dependent variable	$z = f(x, y)$; its graph is a surface	Class 002: E6.1–E11.2
$x,\ y$	Independent variables	Coordinates in the xy-plane	Class 002: all
(x_0, y_0)	Base point	Point where the derivatives are evaluated	Class 002: E9.1–E10.2
$P,\ z_0$	Point on the surface	$P = (x_0, y_0, z_0)$ with $z_0 = f(x_0, y_0)$	Class 002: E9.3
∂	"Partial", read "dou"	Differentiate one variable, hold the others constant	Class 002: E6.1–E6.4
$f_x,\ f_y$	First-order partials	Slopes parallel to the x- and y-axes	Class 002: E6.1, E9.1–E10.2
$f_{xx},\ f_{yy}$	Pure second-order partials	Concavity along x and along y	Class 002: E6.2, E11.1

Symbol	Name	Meaning	Appears in
$f_{xy},\ f_{yx}$	Mixed second-order partials	Change of one slope along the other direction	Class 002: E6.2, E6.4, E11.2
$m,\ n$	Counts	m = number of variables, n = order	Class 002: E6.3
$f'(x_0),\ dy/dx$	Ordinary derivative	Slope of a tangent to a plane curve	Class 002: E7.1, E7.2
∇f	Gradient ("del f")	$f_x\,\mathbf{i} + f_y\,\mathbf{j}$	Class 002: E10.1, E10.2
$\mathbf{i},\ \mathbf{j}$	Unit vectors	Along the x- and y-axes	Class 002: E10.1
$\hat{u} = (a, b)$	Unit direction vector	$a^2 + b^2 = 1$	Class 002: E10.2
$D_{\hat{u}} f$	Directional derivative	Slope of the surface in direction $\hat{u}$	Class 002: E10.2
t	Time (in F14)	Independent variable in the motion analogy	Class 002: E11.3
$z = f(x, y)$	Function of two variables	The quantity being differentiated	Classes 003–004: E14.1, E15.1, E17.1
x, y	Intermediate variables	Depend on t (total derivative) or on u, v (chain rule)	Classes 003–004: E14.1, E17.1
t	Parameter / time	Single final variable in the total derivative; also a new variable in some chain-rule problems	Classes 003–004: E14.1, E14.2
u, v / s, t	New independent variables	Final variables of a composite function	Classes 003–004: E17.1
r, s, t	Named arguments	$r = 2x - 3y$, ... in slide 003·15 (not a radius there)	Classes 003–004: E17.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	Classes 003–004: E17.4, E17.5
$\dfrac{dz}{dt}$	Total derivative	Ordinary derivative when z ends up depending on t alone	Classes 003–004: E14.1
$\dfrac{\partial z}{\partial u}$	Partial derivative of a composite	z depends on u and v	Classes 003–004: E17.1
$dx,\ dy,\ dz$	Differentials	Small (signed) changes; $dz = f_x\,dx + f_y\,dy$	Classes 003–004: E15.1, E15.2

Symbol	Name	Meaning	Appears in
f_x, f_y, f_z	Partial derivatives of f	Kept as symbols when f is unknown	Classes 003–004: E17.1, E18.1, E19.1
$f(x, y) = c$	Level curve / implicit relation	Defines y as a function of x	Classes 003–004: E18.1
$f(x, y, z) = 0$	Level surface	Defines z as a function of x, y	Classes 003–004: E19.1
$f'(r)$	Ordinary derivative of $f(r)$	u depends on x, y, z only through r	Classes 003–004: E17.3
m	Slope	dy/dx at the point; tangent $y - y_0 = m(x - x_0)$	Classes 003–004: E18.2
A, V, h, b, θ	Geometric quantities	Area, volume, height, base, included angle in rate problems	Classes 003–004: E16.1
$u = f(x, y)$	Function	The function tested for homogeneity	Class 005: E20.1, E21.1
n	Degree	Power in $x^n g(y/x)$; any real number	Class 005: E20.1–E23.2
$g(\cdot)$ (topic 1)	Arbitrary function of ratios	$u = x^n g(y/x)$	Class 005: E20.1
$g(u)$ (topic 4)	Extension function	$nf(u)/f'(u)$ (a different g from topic 1)	Class 005: E23.1, E23.2
$f(u)$	Homogeneous function of u	e.g. $\tan u$ when $u = \tan^{-1}(\dots)$	Class 005: E23.1
t	Scale factor	$u(tx, ty) = t^n u$	Class 005: E20.1
u_x, u_y, u_z	First partials	Homogeneous of degree $n - 1$	Class 005: E21.1, E21.2
u_{xx}, u_{xy}, u_{yy}	Second partials	Degree $n - 2$; $u_{xy} = u_{yx}$	Class 005: E22.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	Class 005: E22.3
P, Q	Homogeneous factors	Numerator and denominator of a quotient	Class 005: E20.2
λ	Scale factor	Replaces x, y by $\lambda x, \lambda y$ in the scaling test	Class 006: E25.1

Symbol	Name	Meaning	Appears in
n	Degree	Of u (topic 3) or of $f(u)$ (topic 4)	Class 006: E25.1–E28.2
$f(u)$	Homogeneous function of u	$\sin u$, $\tan u$, e^u, $\log u$	Class 006: E28.1
$f(y/x)$ (slide 3)	Arbitrary function of the ratio	$Z = \sqrt{x}\, f(y/x)$; not the $f(u)$ of topic 4	Class 006: W26.1
$g(u)$	Extension function	$nf(u)/f'(u)$; equals $xu_x + yu_y$	Class 006: E28.1, E28.2
$g'(u)$	Derivative of g	With respect to u	Class 006: E28.2
Z	Auxiliary function	$\cos u$ (slide 3) or $\sin u$ (slide 4, transcript)	Class 006: W26.1, W28.1
z	Third variable	Lowercase, on slide 4 only	Class 006: W28.1
$u_x,\ u_{xy}$	Partial derivatives	$\partial u/\partial x$, $\partial^2 u/\partial x \partial y$	Class 006: E26.1, E27.1
a, (a, b)	Expansion point	Point about which the series is written	Class 007: E30.1, E31.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	Class 007: E31.1–E31.3
$f^{(n)}(a)$	n-th derivative	Printed $f^n(a)$ on slide 007·4; not a power	Class 007: E30.1
$n!$	Factorial	$1 \cdot 2 \cdots n$; divides the n-th order term	Class 007: E30.1–E32.1
$f_x,\ f_y$	First partials	At the expansion point	Class 007: E31.1, E32.1
$f_{xx},\ f_{xy},\ f_{yy}$	Second partials	$f_{xy} = f_{yx}$, hence the 2	Class 007: E31.1, E32.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	Class 007: E31.2
n (in $(1+x)^n$)	Exponent	Any real number; not the order of a derivative	Class 007: E30.4
$L(x, y)$	Taylor polynomial	Slide 007·11's name for the truncated series	Class 007: W31.7
(a, b)	Expansion point	$(1, \frac{\pi}{4})$, $(1, \frac{\pi}{2})$, $(1, 1)$, $(1, -1)$ or $(0, 0)$ in class 008	Class 008: E33.1

Symbol	Name	Meaning	Appears in
$f_x, \ldots, f_{yy}$	Partial derivatives	Evaluated at the expansion point	Class 008: E33.1, E34.1
t	Combination	$2x + y$, $x - y$, $x + 2y$ in the one-variable check	Class 008: E34.2
s	Step along a line	Figure G35.1's parameter only	Class 008: G35.1
$a,\ b$ (in $e^{ax}\sin by$)	Constants	Not the expansion point here (it is the origin)	Class 008: W34.5
(a, b)	Stationary point	Solution of $f_x = f_y = 0$	Class 009: E38.1
$h,\ k$	Small increments	Neighbouring point $(a + h, b + k)$	Class 009: E37.1
Δ	Increment of f	$f(a + h, b + k) - f(a, b)$	Class 009: E37.1
r	f_{xx}	Second partial in x at the point	Class 009: E38.2–E38.4
s	f_{xy}	Mixed partial at the point (not arc length)	Class 009: E38.2–E38.4
t	f_{yy}	Second partial in y (not time)	Class 009: E38.2–E38.4
$rt - s^2$	Hessian determinant	$f_{xx}f_{yy} - f_{xy}^2$	Class 009: E38.3
$p,\ q$	First partials	$p = f_x$, $q = f_y$ (lecture T010b)	Class 010: E39.1
$r,\ s,\ t$	Second partials	f_{xx}, f_{xy}, f_{yy} at a point	Class 010: E39.2, E39.3
$rt - s^2$	Hessian	Decides extremum or saddle	Class 010: E39.2, E39.3
T	Temperature	$400xyz^2$ or $kxyz^2$; not the second partial t	Class 010: E41.1
k	Constant	Positive constant in $T = kxyz^2$	Class 010: E41.1
a	Radius	Sphere $x^2 + y^2 + z^2 = a^2$	Class 010: E41.1
A	Surface area	Metal used by the open box	Class 010: E41.2
V	Volume	Fixed capacity $xyz = V$ (10 m³)	Class 010: E41.2
$f(x, y, z)$	Objective	Function to maximise or minimise	Class 011: E42.1
$\phi(x, y, z)$	Constraint	$\phi = 0$; printed $\varnothing$ on the slides	Class 011: E42.1, E42.2

Symbol	Name	Meaning	Appears in
λ	Lagrange multiplier	Undetermined constant; not in the answer	Class 011: E42.1–E42.3
F	Auxiliary function	$f + \lambda\phi$	Class 011: E42.1
r	Radius	Of the sphere $x^2 + y^2 + z^2 = r^2$; not f_{xx}	Class 011: E43.1
$a,\ b,\ c$	Semi-axes	Of the ellipsoid (in W44.4: positive constants)	Class 011: E43.1, W44.4
$2x,\ 2y,\ 2z$	Edges	Of the inscribed box; volume $8xyz$	Class 011: E43.1

Remark. **Class 001.** The slides write z_x and f_x for $\frac{\partial z}{\partial x}$; the lecture says the subscript form is the one used in problems. In the subscript notation the subscripts are read left to right in the order of differentiation; in the ∂ notation the order is read right to left (slide 001·17).

Class 002. No physical units here: every slope is a pure number (rise per unit run) unless x, y, z carry units. Then f_x has units of z per unit of x.

Classes 003–004. Watch the letters that change meaning between problems: t is the parameter in the total derivative but one of the new variables in the chain-rule problems (s, t); r, s, t are intermediate variables in slide 003·15 but r is a radius elsewhere; f is the outer function in composite problems and the implicit relation $f(x, y) = 0$ in class 004.

Class 005. The letter g has two meanings: the arbitrary function of ratios in $x^n g(y/x)$ (topic 1), and $g(u) = nf(u)/f'(u)$ in the extension (topic 4).

Class 006. Two clashes come from the slides themselves. On slide 006·4 lowercase z is the third variable while the lecture writes the auxiliary function as uppercase $Z = \sin u$; on slide 006·3 $Z = \cos u$ is the auxiliary function. And f is both the homogeneous function of u ($f(u) = \sin u$, $\log u$, e^u) and, on slide 006·3, the arbitrary function in $\sqrt{x}\, f(y/x)$.

Class 007. $h = x - a$ and $k = y - b$ are the shorthand used in the worked examples and are added for clarity; the slides write $(x - a)$ and $(y - b)$ in full. $f^{(n)}$ (the n-th derivative) is printed as f^n on slide 007·4; it is not a power.

Class 008. $h = x - a$ and $k = y - b$ are shorthand added for clarity; the slides write $(x - 1)$, $(y - \frac{\pi}{4})$ and so on in full. "Up to three terms" (slide 008·5) means three non-zero terms, as the lecture explains; it is not the same as "up to third degree".

Class 009. The deck uses $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$; some books write A, B, C or $D = f_{xx}f_{yy} - f_{xy}^2$. The t here is a second partial derivative, not time; the s is not arc length.

Class 010. The slides write $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ and test the sign of $rt - s^2$; other books call it the discriminant D or the Hessian. The lecture of T010b also names the first partials $p = f_x$ and $q = f_y$. Do not confuse $t = f_{yy}$ with the temperature T of slides 010·3 and 010·9.

Class 011. The slides write the constraint as $\phi(x, y, z) = 0$ (also printed as $\varnothing$) and the auxiliary function as $F = f + \lambda\phi$; the lecture also calls it the Lagrangian. λ is the Lagrange multiplier; its value is found but never needed in the answer. In the sphere problems r is the radius of the sphere, not the second partial f_{xx} of class 009.